BIO204 1st Mid semester Exam (2026) IISER Mohali

Date: 06/02/2026

Duration: 60 min

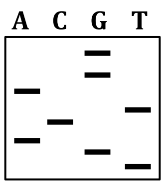
Maximum marks: 30

Instructions:

a. All questions are compulsory. Each question in Sections A and B carries 1 mark.

Sections A

1. Which description is best for MeSH (Medical Subject Headings) term?
   a. A free, online medical encyclopedia for patients to search medical terms.
   b. Serve as a standardized vocabulary for indexing and cataloging biomedical literature.
   c. Serve as a database for commonly occurring words in medical literature.
   d. To list the common names of medical disease, drugs, description in different countries.
2. An integrated search and retrieval system provided for NCBI database is called as:
   a. Pubmed
   b. Algol
   c. Entrez
   d. RefExtract
3. Which is a composite database:
   a. GenBank
   b. PDBj
   c. TAIR
   d. Uniprot
4. A database documenting protein motifs is:
   a. KEGG
   b. OMIM
   c. GEO
   d. PROSITE
5. From the Sanger sequencing gel provided below, the template DNA sequence is:



   a. 5’-TGACTAGG-3’
   b. 5’-GGATCAGT-3’
   c. 5’-CCTAGTCA-3’
   d. 5’-ACTGATCC-3’
6. In genome sequencing, DNA is fragmented and sequenced; the number of times a nucleotide is sequenced is known as:
   a. Read depth
   b. Phred score
   c. HiFi read
   d. Read quantity
7. Method used for structure determination of large protein complexes is:
   a. NMR
   b. X-ray crystallography
   c. Electron microscopy
   d. Cryo-electron microscopy
8. Program for *de novo* assembly of genome using *de bruijn graph* is:
   a. Canu
   b. denoCanu
   c. Arachne
   d. SOAPdenovo
9. Protein sequence motifs are usually identified in:
   a. Multiple sequence alignment
   b. Pairwise sequence alignment
   c. Multiple DNA sequence alignment of gene.
   d. Alignment of multiple motifs
10. Which structure determination method directly determines the atomic coordinates of a protein?
   a. NMR
   b. X-ray crystallography
   c. Electron microscopy
   d. Cryo-electron microscopy

Section B

11. Chaos Game Representation of 'TTTTTTTTTTT' DNA sequence will be seen as ***diagonal line from center to corner depicting T nucleotide***.
12. The Information content of DNA sequence 'AAAAAA' is ***2 bits***.
13. Three ontologies or aspects defined for GO are ***Biological Process, Molecular Function****, and* ***Cellular Component***.
14. For DNA 'GCGCTATAGCGGCCTAATTT' the GC% is ***50%***.
15. Assuming linear gap penalty of -2, match =+3 and mismatch =-1, the score of the alignment given below is ***4***.

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Seq1: KPKKAKSAPAPA
Seq2: KP--VRT-PA-A
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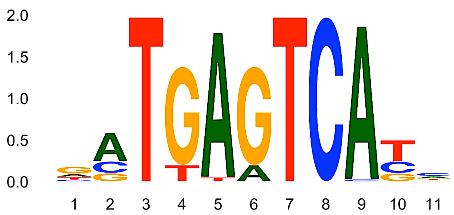
16. Based on the provided sequence alignment, sequence identity ***41.7%*** and sequence similarity ***66%*** *(assuming hydrophobic, polar and charged substitution to be similar)***.**
17. Mr. X sequenced parakeet genome sequence and decided to submit the assembled genome sequence. Mr. X will submit it to the ***GenBank/DDBJ/ENA*** database (name) and the translated gene sequences will be deposited in ***Uniprot*** database (name).
18. In PAM250, the number 250 refers to ***evolutionary distance in PAM units (1 PAM unit is 1% of amino acid sequence has accepted mutations) OR 250 mutations per 100 amino acids during evolution***.
19. In BLOSUM45, the number 45 refers to ***sequence identity threshold of 45% used for clustering sequence blocks***.
20. Motifs are defined as ***short conserved regions of DNA or protein sequences that usually are associated with biological function***.
21. What is probability of protein sequence S='GKTGTT' assuming 0^{th} order Markov chain? Assume equal probable occurrence of amino acid sequence. **$(1/20)^6$**
22. In genome assembly, a contig ***is a set of DNA segments overlapping to provide contiguous representation of a genomic region***, while a scaffold ***is a set of contigs oriented and positioned relative to each other, often separated by gaps***.

Section C

1. Based on the sequence logos given below, the most conserved positions are ***3, 7, 8*** and the least conserved position is ***11***. **(2 marks)**



2. A new type of sequence consists of 8 characters (B, C, D, E, F, G, H, I). What is the Shannon entropy assuming equal probability of all characters (null distribution), and what is the entropy for a sequence composed of a single repeated character? **(2 marks)**
3 bits and 0 bits

3. A DNA sequence from a newly discovered archaeal species is determined, and initial evidence suggests it may contain TF binding site for ZnBS transcription factor. The binding motif has been modeled in the position weight matrix (PWM) given below. **(4 marks)**
   a. Determine the consensus sequence of the binding motif from the PWM.
      GTAA
   b. Find the most likely starting position of this motif in the target sequence S='AGTATG'. Justify.
      The matching score after scanning PWM on sequence S starting from position 1: -0.5
      The matching score after scanning PWM on sequence S starting from position 2: 3.65
      The matching score after scanning PWM on sequence S starting from position 3: -0.4
      The most likely motif starting position is position 2.

PWM (motif)

	0	1	2	3	4
A	0.25	-1.25	0.35	1.20	1.50
C	0.25	0.20	-1.25	-1.40	0.45
G	0.25	1.25	-1.00	0.25	-1.25
T	0.25	0.25	0.70	0.25	0.50